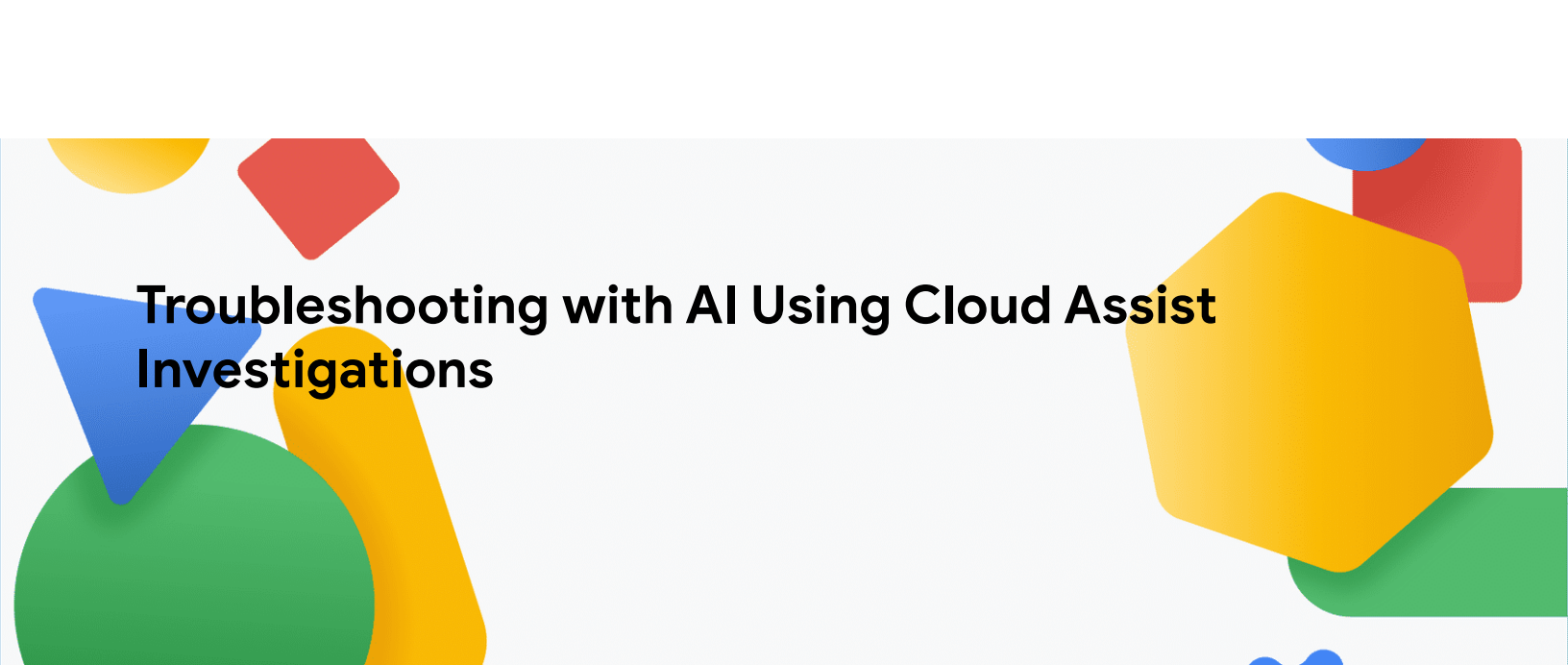




Troubleshooting with AI Using Cloud Assist Investigations



Cloud Assist Investigations is an AI-enabled tool for iterative troubleshooting in Google Cloud environments. It aims to help users troubleshoot common issues to find the root cause of problems that may occur when using Google Cloud services.

This course equips DevOps, SRE, and platform engineers with the skills to effectively use Cloud Assist Investigations for streamlined troubleshooting of Google Cloud services. The course focuses on foundational knowledge of Cloud Assist Investigations, with the intent to reduce troubleshooting time and enhance overall operational efficiency for users of Google Cloud.

Click the **Start Course** button to begin.

INTRODUCTION TO CLOUD ASSIST INVESTIGATIONS



What's in it for me?



Gemini Cloud Assist



Cloud Assist Investigations overview

CLOUD ASSIST INVESTIGATIONS INTERFACE AND CORE CONCEPTS

 Core concepts

 Create and review investigations

CONDUCT INVESTIGATIONS AND PERFORM DIAGNOSTICS

 Investigate and diagnose issues

 What did I walk away with?

What's in it for me?

Cloud Assist Investigations is a root-cause analysis (RCA) tool for troubleshooting your infrastructure and applications in complex and distributed cloud environments.

Cloud Assist Investigations can help you understand, diagnose, and resolve issues on Google Cloud. With Cloud Assist Investigations, you can streamline incident responses, reducing the time to resolution and improving your overall service availability.

Cloud Assist Investigations is a feature of Gemini Cloud Assist, an AI-powered product that helps you design, optimize, and operate applications in Google Cloud. We'll briefly review Gemini Cloud Assist at the beginning of this course.

By the end of this course, you will be able to:



- ✓ Access and navigate the Cloud Assist Investigations interface.
- ✓ Utilize core features of Cloud Assist Investigations to diagnose common issues.
- ✓ Troubleshoot incidents with Cloud Assist Investigations.

Gemini Cloud Assist

Cloud Assist Investigations is a feature of [Gemini Cloud Assist](#), a product that offers AI-powered assistance to help you simplify the application lifecycle on Google Cloud.

Types of generative AI assistance

Gemini Cloud Assist provides the following types of gen AI assistance to build and design applications. Click each tab to learn more.

DESIGN AND BUILD

DIAGNOSE AND RESOLVE

OPTIMIZE COST AND USAGE

- Design, build, and manage applications with natural language interaction in the [Application Design Center](#) in the Google Cloud console.
- Generate Google Cloud CLI commands and Terraform code for infrastructure and organizational policies.

DESIGN AND BUILD	DIAGNOSE AND RESOLVE	OPTIMIZE COST AND USAGE
• Diagnose issues based on logs, errors, or natural language prompts using AI-driven Cloud Assist Investigations. • Use AI-powered workflows to troubleshoot and resolve database performance issues.		

DESIGN AND BUILD	DIAGNOSE AND RESOLVE	OPTIMIZE COST AND USAGE
• Optimize cloud spend by analyzing costs, trends, waste, and utilization in FinOps Hub. • Query project costs and billing information with natural language in Cloud Hub. • Optimize database costs and generate savings by configuring instances based on usage patterns in Database Center.		

You can also get gen AI assistance for your Google Cloud resources with Gemini Cloud Assist chat, Google Cloud Observability, Firebase, and other tools. For more information, visit the [documentation](#).

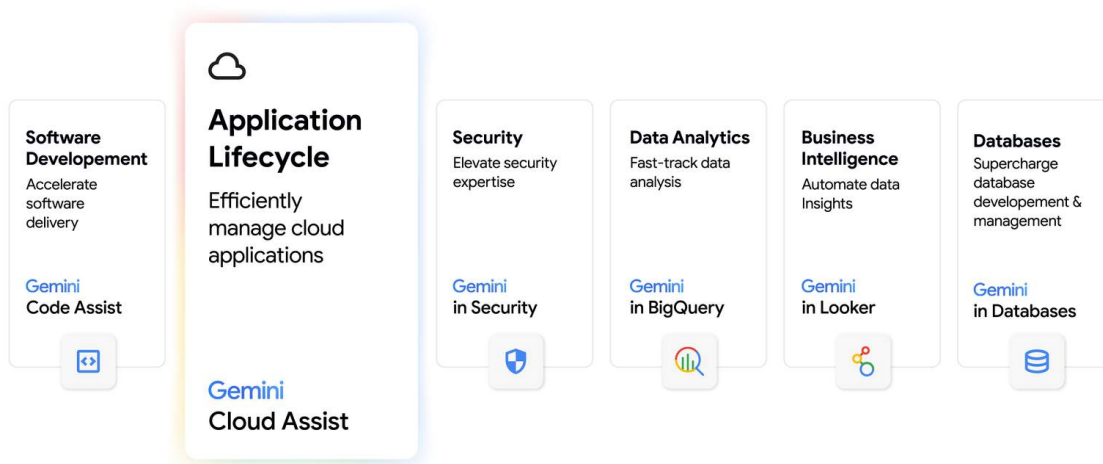
Interacting with Gemini Cloud Assist

You can interact with Gemini Cloud Assist in multiple ways:

- [Cloud Assist panel](#): Get conversational assistance using this panel in the Google Cloud console and the Cloud Mobile app.
- [Cloud Hub](#): Optimize resource and application costs and utilization in Cloud Hub in the Google Cloud console.
- [Database Center](#): Learn more about your database fleet health by prompting Gemini to answer questions about fleet health, inventory, and database configuration information.
- [Application Design Center](#): Design optimal cloud applications that can be managed throughout their lifecycle with Gemini in the App Design Center.
- [Cloud Assist Investigations](#): Analyze, troubleshoot, and diagnose technical issues within the Google Cloud console using Cloud Assist Investigations.

Gemini Cloud Assist is part of the Gemini for Google Cloud portfolio of products and services.

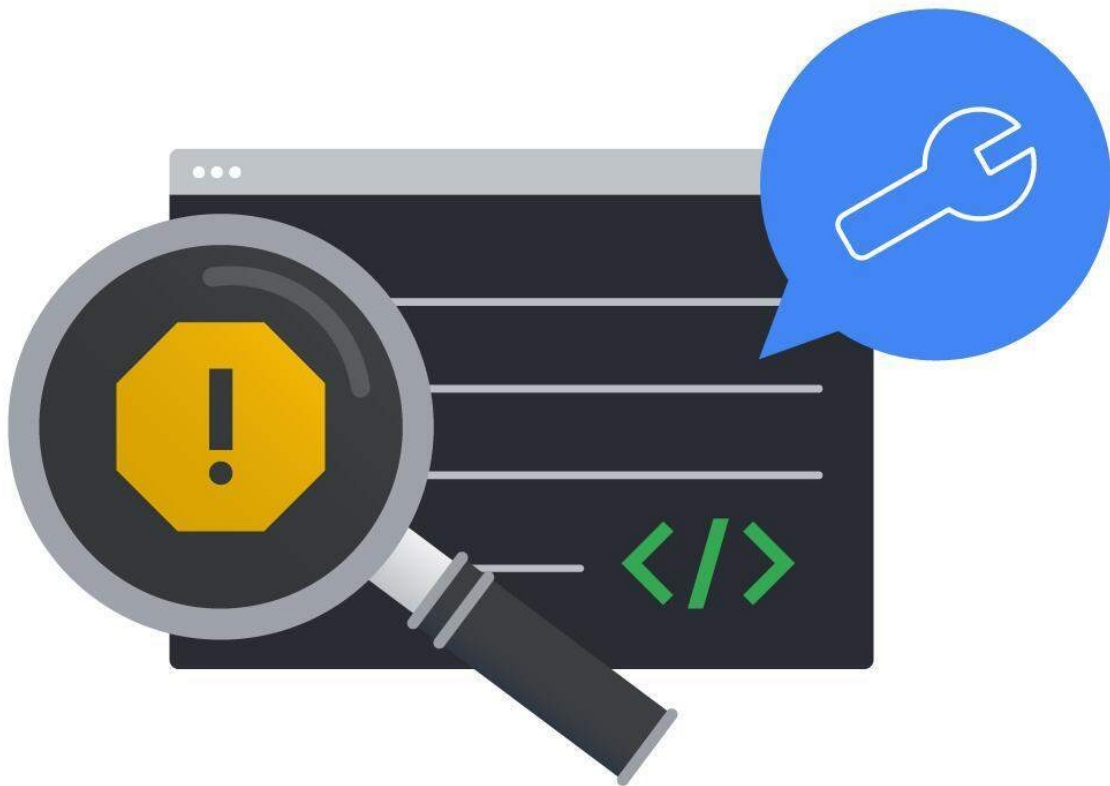
Gemini for Google Cloud Portfolio



Cloud Assist Investigations overview

The need for speed in troubleshooting

When things break and an outage occurs, organizations move into firefighting mode to identify and fix the issues that have caused the outage. Outages can be very costly, so fixing the issue quickly and minimizing the impact are incredibly important.



Mean time to recovery (MTTR)

- A key metric used to measure the efficiency of incident responses
- Average time taken to restore a service to full operation after a failure

To reduce the MTTR and improve overall service availability, it is important to streamline incident responses.

Cloud Assist Investigations

With Cloud Assist Investigations, you can streamline incident responses by reducing the time to resolution and recovery of your service on Google Cloud.

Click each button below to learn more about the benefits of Cloud Assist Investigations.

Integrate your current workflows

- Cloud Assist Investigations is integrated into your current workflows, so you can troubleshoot issues in context.
- You can trigger investigations from various products and services you use with your application on Google Cloud, such as Logs Explorer.

Understand issues —

- Cloud Assist Investigations provides observations and insights on the state of your Google Cloud environment.
- Data sources, such as logs, configurations, and runbooks are analyzed to help you focus on information most relevant to your issue.

Diagnose issues —

- Cloud Assist Investigations synthesizes observations and uses domain-specific knowledge to identify the probable root cause(s) of an issue.
- To explain the context around the root cause, public and private knowledge is used.

Resolve issues —

- To help resolve an issue, Cloud Assist Investigations recommends troubleshooting steps or fixes.
- References to specific resources or time periods provide relevant information and help resolve the issue faster and with less effort.

Get additional support —

- For additional support, your investigation is seamlessly transferred into a Google Cloud support case.

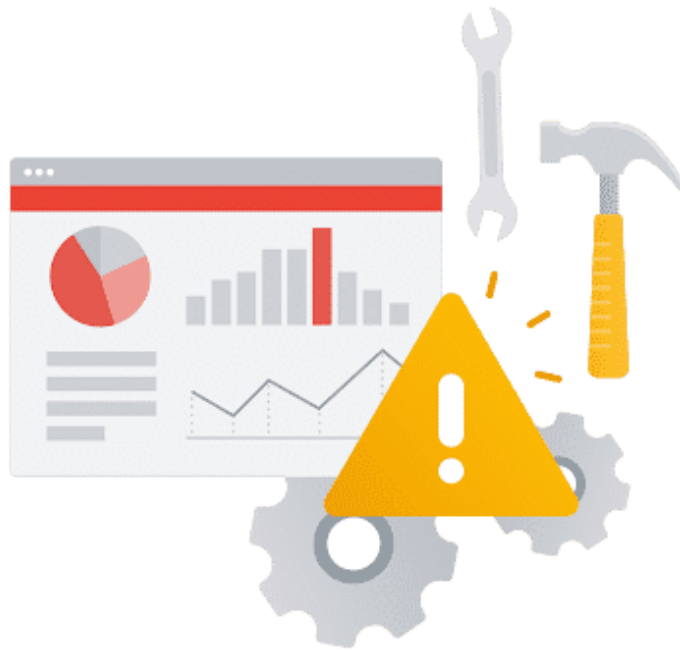
- Investigation details with additional context are used by a support engineer who can help resolve the case faster.

Troubleshoot issues in context

Troubleshoot issues within the context of your current workflow by triggering investigations from:

- Log messages with a severity of "Warning and higher" in Logs Explorer
- [Product pages](#) such as Google Kubernetes Engine
- Health & troubleshooting page in Cloud Hub
- Main Investigations page in the Google Cloud console
- Gemini Cloud Assist chat panel
- Cloud Monitoring alerts

You can also interact and ask questions about your investigations in chat.



Knowledge check

Which two statements describe MTTR?

- ☐ The rate per minute at which incidents occur with a service.
- ☐ A key metric used to measure the efficiency of incident responses.
- ☐ Average time taken to restore a service to full operation after a failure.

SUBMIT

Core concepts

Let's discuss the core concepts and terminology used in Cloud Assist Investigations.

Concepts

When working with Cloud Assist Investigations, you start an investigation on an issue that may be reported, for example, in Logs Explorer, in the Google Cloud console. Observations are then produced based on the analysis of the issue, and hypotheses about the cause of the issue are generated.

Click each flashcard to learn more.

Issue

An issue is information that can include errors, a description, a timeline, or the affected resources.

Observation

An observation represents the insights about your environment's state that are most relevant to the issue.

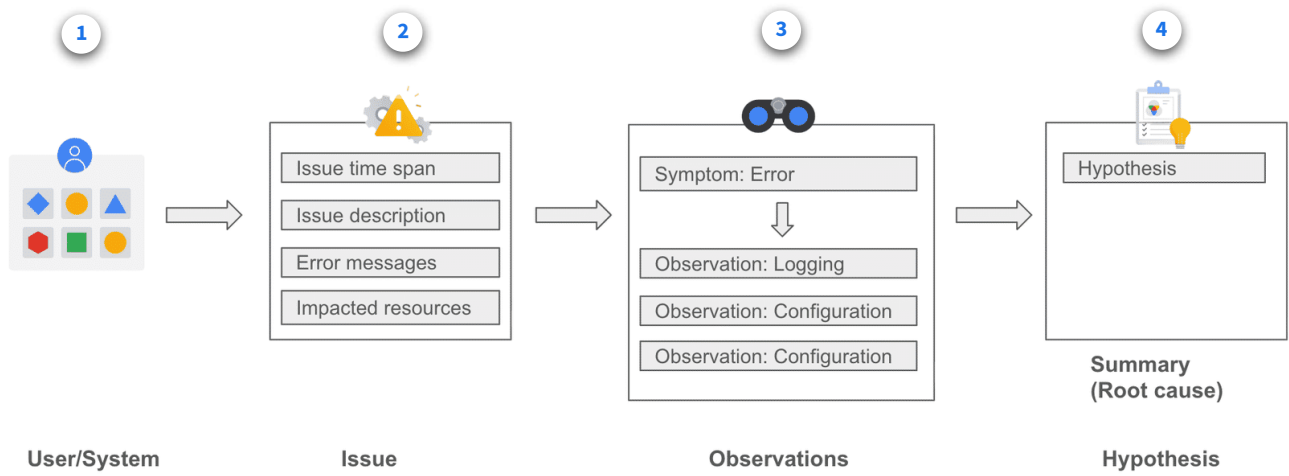
Hypothesis

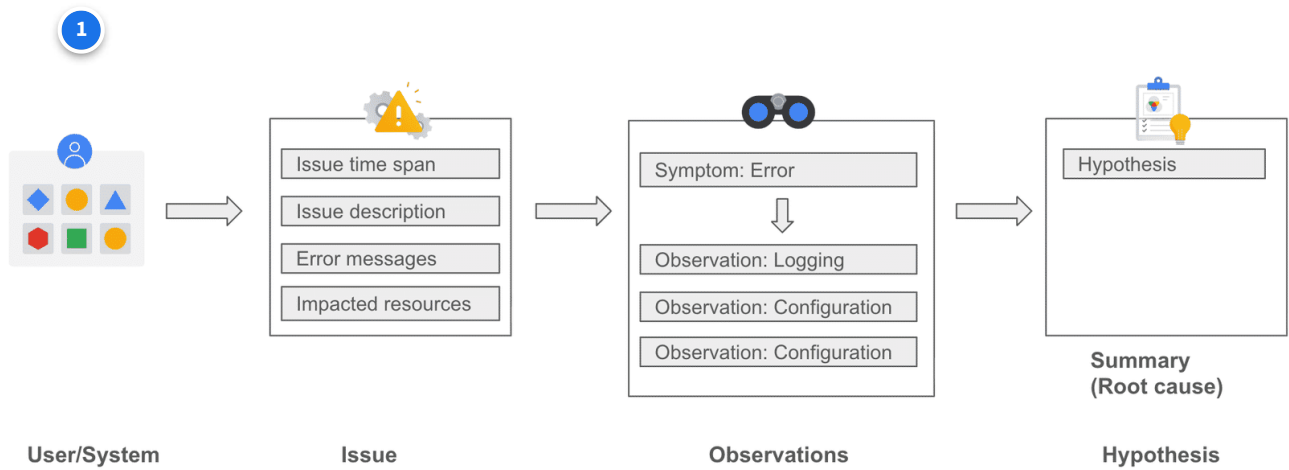
A hypothesis synthesizes the observations and uses domain-specific knowledge to identify probable root causes for the issue.

Investigations workflow

Here is a typical workflow when using Cloud Assist Investigations.

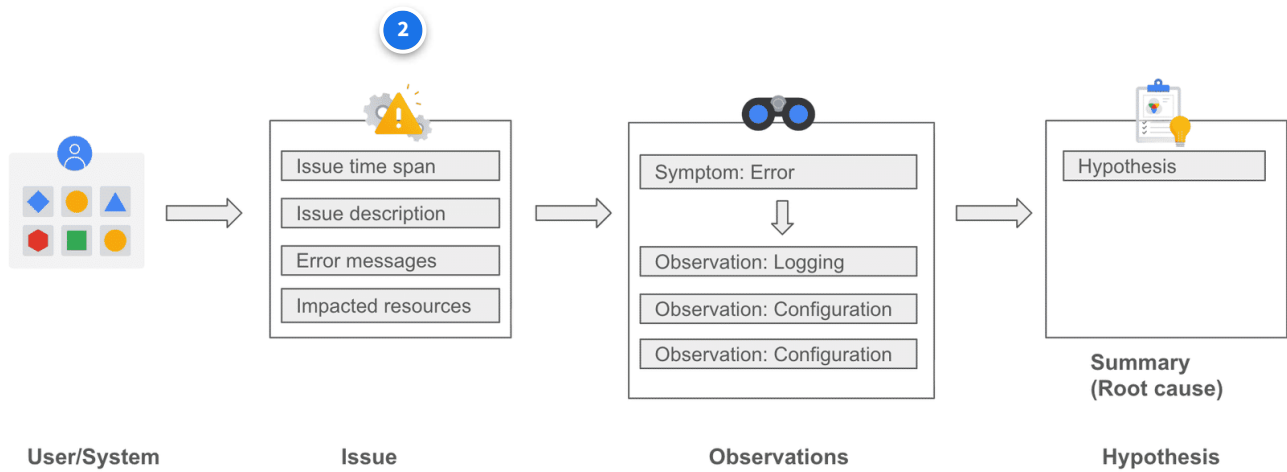
Click each marker to learn more.





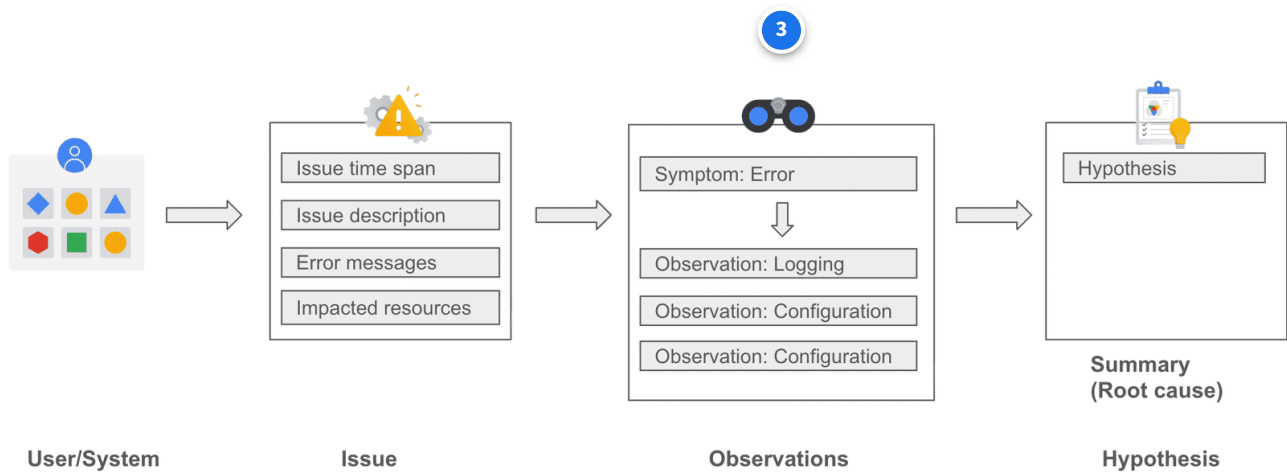
User or system

A user or system interacts with your application resources and generates an issue.



Issue

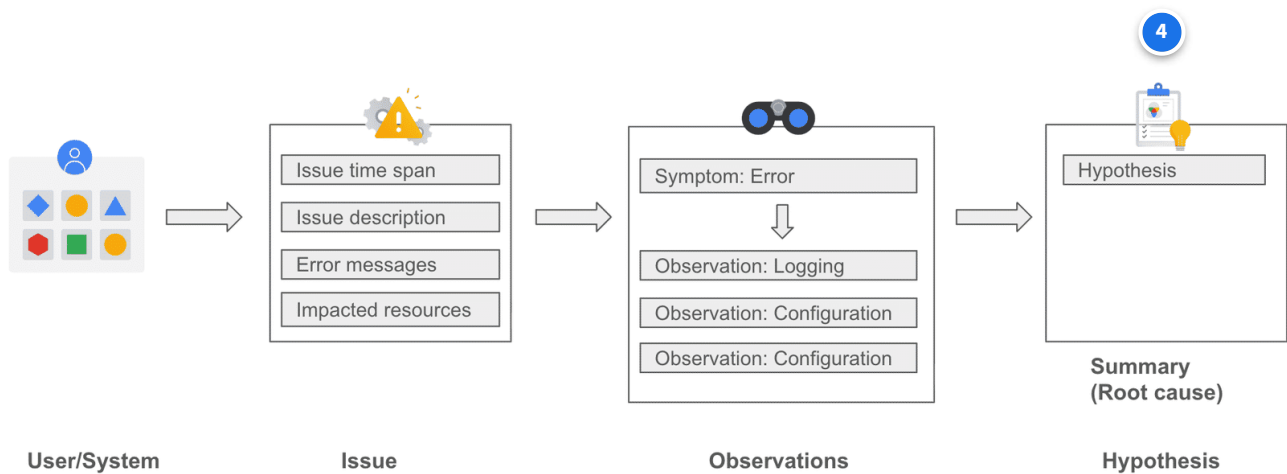
The issue is provided to Cloud Assist Investigations and can include error messages, a description, a timeline, or the resources that are impacted.



Observations

Observations that are most relevant to the issue are then generated. They contain insights about your environment's state and help you to quickly understand what is happening in your environment. The context of your Google Cloud environment is used to analyze data sources such as logs, configurations, and metrics.

Observations are then ranked and filtered to help you focus on the information that is most relevant to your issue.



Hypotheses

Investigations synthesize the observations and uses domain-specific knowledge to identify probable root causes that are provided as hypotheses.

You can iterate on an investigation and create new revisions to drill deeper into certain areas.

Watch this video that shows how you can use Gemini Cloud Assist Investigations to troubleshoot issues with your workloads and resources on Google Cloud.

Click the play button to watch the video.

Use Gemini Cloud Assist to inve...



Transcript

0:00

[MUSIC PLAYING]

0:020:07

KYLE BENSON: Hi.

0:08

My name is Kyle Benson.

0:09

I'm a product manager at Google Cloud.

0:12

Good observability and well-honed troubleshooting

0:14

skills are fundamental to any on-call engineer's success.

0:19

They provide the foundational data and tools

0:21

that we need as humans to think critically about problems.

0:24

But, sometimes, the specter of unknown unknowns

0:27

can make troubleshooting difficult or feel impossible.

Overview

0:31

We're trying to help make this process easier for customers

0:34

of Google Cloud with our new Investigations tool, which

0:37

is part of Gemini Cloud Assist.

0:40

The Investigations feature is designed

0:42

to aid in the troubleshooting flow started

0:44

by you and your teammates, so you

0:46

can start from scratch or within context across Google Cloud.

0:50

The Investigations feature takes the symptoms or error messages

0:54

you provide, identifies the underlying issues,

0:57

and recommends actionable solutions.

0:59

Let's see how we can create an investigation

1:01

to troubleshoot a live issue in our environment.

1:04

Create an investigation

1:10

For this example, we have a sample workload running

1:13

in Google Kubernetes Engine.

1:15

So we've received a page notifying us that there's

1:17

an issue with our service.

1:20

I'm going to open the Google Kubernetes Workloads

1:23

page to take a look at my application.

1:27

It looks like the inventory app is having some issues.

1:31

I can see that there's a CrashLoopBackOff error,

1:33

and it doesn't have the minimum availability.

1:36

To dig into this further, I'm going

1:39

to go into the container logs for this service

1:41

and see what's happening.

1:44

Once Cloud Logging loads these logs,

1:47

I can see that there are some intermittent errors related

1:50

to some kind of failure.

1:52

If I click on this most recent error,

1:55

we can see that Gemini offers this new Investigate button.

1:58

And if I click through, it'll let

2:01

me create a new investigation.

2:04

I'm going to change the name here to System is down.

2:08

And just to show how important and urgent this is,

2:11

I'll add a couple of exclamation points.

2:13

I'm going to leave the start time.

2:15

And then I'm going to scroll down here a little

2:17

to say what type of issue am I facing.

2:20

I can see "I have an error message"

2:22

or "I want to describe what happened."

2:24

In this case, we have an error message

2:25

we think might be related to this failure.

2:28

So I'm going to leave it as is with everything that it

2:31

populated.

2:33

Then I'm going to let the relevant resources stay

2:36

the same as well, and I'm going to click Create.

2:382:41

Once this investigation is created,

2:44

Gemini will analyze the relevant resources

2:47

and any of the things it thinks might

2:48

be relevant to try and solve this issue.

2:51

So we'll give it a few moments to let it complete

2:54

and see what's going on.

2:55:00

It looks like it completed.

3:02

We can first see a few relevant observations.

3:05

So if we wanted to see what was happening here, things that

3:08

might be related to what created this issue,

3:10

we can see that from here.

3:12

We can also go straight to the summary of findings.

3:15

It looks like the application recognizes some error on line 14

3:20

within the Python code.

3:21

What this tells me, as the on-call engineer,

3:24

is that there is some kind of error that

3:26

was pushed to a recent change in this service

3:28

and that the team responsible for this app

3:30

needs to update the code to more gracefully solve these errors so

3:34

that the container doesn't spontaneously crash

3:36

when it hits that problem.

3:38

From here, what we need to do is have the on-call engineer

3:42

contact the other teams, reroll the package,

3:45

and reroll our deployment.

3:46

This should get us back online.

3:48

If you try the suggestions provided by the Investigations

3:51

feature and the issue is still unresolved,

3:53

you can always create a case with Google Cloud Support

3:56

and pass along the steps you've tried already with Gemini.

4:00

In addition to the Logs Explorer,

4:02

there are ways to initiate investigations

4:04

across Google Cloud, such as from the Cloud Console

4:08

chat or other services we're going

4:10

to be introducing soon as well.

4:12

[MUSICAL FLOURISH]

Learn more

4:14

Investigations redefines troubleshooting, leveraging

4:17

generative AI to instantly surface insights, guide

4:20

your decisions, and help you resolve issues

4:22

faster than ever before.

4:25

To learn more about investigations

4:26

and other features of Gemini Cloud Assist, visit our website.

4:30

Or you can go to Google Cloud console

4:32

to enable Gemini Cloud Assist today.

4:34

I hope you found this helpful, and we're

4:36

looking forward to getting your feedback on Gemini Cloud Assist.

4:39

[MUSIC PLAYING]

4:42

You can resolve the issue with recommended fixes or additional troubleshooting steps provided by Cloud Assist Investigations.

Knowledge check

Match each concept with its definition.

⋮ Investigation	The main workflow used to investigate and diagnose an issue.
⋮ Hypothesis	Represents the insights about your environment's state that are most relevant to the issue.
⋮ Observation	Information that can include errors, a description, a timeline, or the affected resources.
⋮ Issue	Synthesizes the observations and uses domain-specific knowledge to identify probable root causes for the issue.

SUBMIT

Create and review investigations

In this lesson, we'll review the steps to create and use investigations. As mentioned earlier, there are multiple entry points where you can initiate an investigation with Cloud Assist Investigations.

The entry points in the Google Cloud console are on:

- The Investigations page
- Logs Explorer
- The alerting page in Cloud Monitoring
- The Gemini Cloud Assist chat panel
- Cloud Hub
- Supported Google Cloud product pages

You can also initiate and view investigations in the [Google Cloud mobile app](#) by navigating to the Observability tab and selecting Investigations.

Note: Additional entry points to access Cloud Assist Investigations are being added.

The Investigations page

You can access this page by navigating to Cloud Assist Investigations in the Tools section of the products page in the Google Cloud console.

All products

Jump Start Solutions

Solution deployments

Categories

Management

Compute

Storage

Analytics

Networking

Distributed Cloud

Serverless

Databases

Observability

Operations

Security

App Development

CI/CD

Artificial Intelligence

Integration Services

Tools

Tools

Discover productivity resources for your cloud

Name	Description
Identity Platform	Google-grade identity and access management
Deployment Manager	Templated infrastructure deployment
Service Catalog	Internal solutions catalog
Active Assist	Proactively optimize your cloud environment
Carbon Footprint	Your cloud carbon emissions
Splunk Cloud	Data-to-Value Platform
Cloud Assist Investigations	Troubleshoot issues quickly with Cloud Assist Quickly analyze, troubleshoot, and diagnose technical issues to find root causes, reducing cloud downtime and operational friction.
Cloud Workstations	Development environments in the cloud
Infrastructure Manager	Infrastructure as Code (IaC) deployments
Migration Center	Move on-premises workloads to Google Cloud
Service Health	Disruptive events impacting Google Cloud

Access Cloud Assist Investigations from the Google Cloud console

Introduction

Let's review the process to create an investigation from the Investigations page in the Google Cloud console.

Click the **Start** or **Next** button (>) to learn more.

Step 2

Prerequisites

Before you can create an investigation, you must have the following prerequisites:

1. [Required APIs](#) enabled for your Google Cloud project.
2. Required permissions to create an investigation. You can acquire these permissions by having the *Investigation Creator* IAM role on your project.
 - When you create an investigation, you're granted the *Investigation Owner* IAM role on that investigation, which enables you to review the results of the investigation.
3. Access to the Google Cloud product being investigated.
 - The investigation's access to the product's APIs and other information is limited to the access you have, based on your credentials. (We'll review this topic in more detail later).
4. If you're investigating an application, the application must be contained within [App Hub](#).

Step 3

Create a new investigation

Cloud Assist Investigations / Create investigation

Create new investigation

Preview

Investigate common issues efficiently for initial hypotheses. May not find rare problems. Our [supported product list](#) is growing.

Name

New investigation - Aug 15, 2025, 11:04:04 PM

Describe the issue

Provide details about the problem, paste the exact error message, or both

Issue description *

Start time *

8/15/25, 10:34 PM PDT

End time (optional) PST

Defaults to 30 minutes prior, max 30 days ago

Time issue ended

Select resources

Add resources that are affected by the issue to help Cloud Assist focus its investigation. Cloud Assist Investigations currently [supports key Google Cloud products](#). Unsupported resources are not listed and require manual addition.

Target project *

apigee-edge-hm

Select

+ Add resource

Cloud Assist Investigations can generate inaccurate responses. Validate responses for accuracy. [Learn more about Gemini for Google Cloud and responsible AI](#)

Create

Cancel

To create a new investigation, click the Cloud Assist Investigations link on the Products page in the Cloud console. Then, click **Create** on the Cloud Assist Investigations home page.

This opens the new investigation form, where you provide the following information:

- **Investigation name:** This helps to identify the investigation in your history (optional).
- **Description:** Describe the issue in natural language or with a verbatim error message.
- **Start time:** The time when the issue began.
- **End time:** The time when the issue ended (optional).
- **Project or Application:** To target or scope the investigation.
- **Resources:** The resources of supported Google Cloud products in scope that are affected by the issue. Optional if errors messages are provided, but still recommended. For example, a Cloud Storage bucket.

After providing these details, begin the investigation by clicking **Create**.

Step 4

Review an investigation

The screenshot displays the AWS IAM console interface for reviewing an investigation. At the top, a header bar shows the issue title "Issue: New investigation - Jul 16, 2025, 1:14:26 PM" and a "Complete" status with a dropdown arrow. Below this, a section titled "2 relevant observations" with an "Expand all" link contains two items: "Bucket Access Restriction due to IAM Configuration" (with a "Config analysis" tag and "2 hours ago" timestamp) and "Cloud Storage Access Logs Show No Errors" (with a "Logs" tag). The "Hypotheses" section, also expandable, states "We have one potential hypothesis for the issue you've described" and lists a single hypothesis: "1 Bucket Access Restricted by Public Access Prevention and IAM Configuration". This hypothesis includes an "Overview" section explaining that the Cloud Storage bucket 'clsaitools' is inaccessible due to a restrictive configuration involving enforced public access prevention, bucket policy, and IAM policies. It also includes a "Recommended fixes" section with a bullet point: "Review IAM Permissions: Ensure that the appropriate service accounts or users have the necessary IAM roles to access the bucket and its contents."

Issue: New investigation - Jul 16, 2025, 1:14:26 PM Complete

2 relevant observations Expand all

Bucket Access Restriction due to IAM Configuration Config analysis 2 hours ago

Cloud Storage Access Logs Show No Errors Logs

Hypotheses Expand

We have one potential hypothesis for the issue you've described

1 Bucket Access Restricted by Public Access Prevention and IAM Configuration Expand

Overview:

The Cloud Storage bucket 'clsaitools' is inaccessible due to a restrictive configuration that includes enforced public access prevention, bucket policy only, and uniform bucket-level access. These settings, in combination, ensure that access to the bucket is exclusively controlled by IAM policies, effectively blocking any access not explicitly granted through IAM. Additionally, a soft delete policy is in effect, retaining deleted objects for 7 days, which could contribute to the perception of inaccessibility if objects were recently removed.

Recommended fixes:

- Review IAM Permissions: Ensure that the appropriate service accounts or users have the necessary IAM roles to access the bucket and its contents.

After you create an investigation, it can take several minutes to complete.

Once completed, you can view the details of the investigation from the same entry point or from your project's investigation history in the console.

The image is an example of the investigation details of an issue related to accessing a Cloud Storage bucket. We'll review these details in the next module.

Summary

The investigation details contains sections that include information about the issue, the observations, and the hypotheses.

The details include recommended fixes to help you correct the issue, or include additional troubleshooting steps to confirm or refute the hypotheses.

Logs Explorer

The Logs Explorer in the Google Cloud console is another entry point from where you can initiate an investigation.

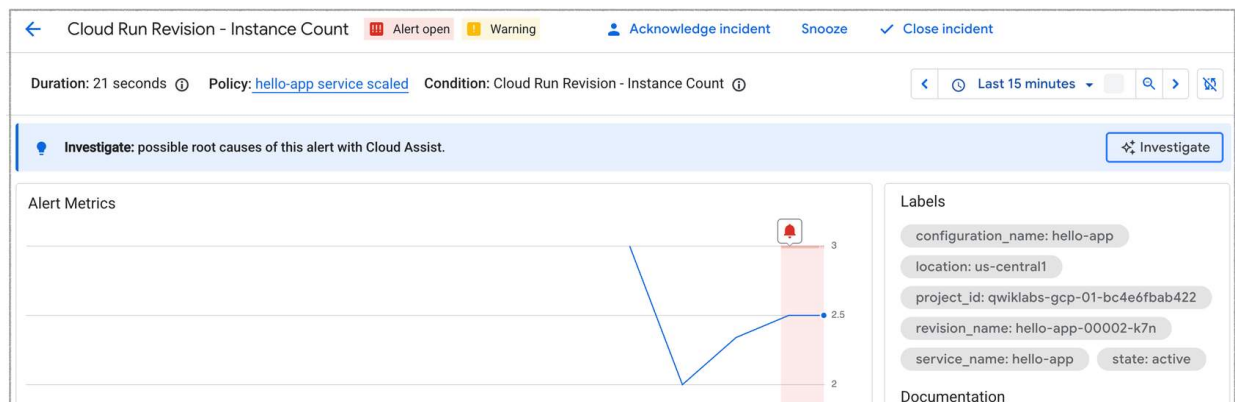
- The **Investigate** button appears in the expanded log message for supported resources when the log message has a severity level of "Warning" or higher.
- When initiating an investigation in Logs Explorer, the log message, start time, and any relevant resources from the log are automatically prepopulated into the investigation creation form.
- You can optionally edit or add information before running the investigation.
- Based on your Google Cloud entitlements, if Cloud Assist Investigations is unavailable, the **Investigate** button provides an explanation of the log entry in the Gemini Cloud Assist chat panel.

Cloud Monitoring alerting

Cloud Monitoring provides different services that you can use to understand the health and performance of your applications and the other Google Cloud services that you use.

The [alerting feature of Cloud Monitoring](#) notifies you when the value of a performance metric meets criteria that you define.

- The alerting process helps you respond to issues when the performance of an application fails to meet acceptable values.
- The **Investigate** button appears for alerts that have a resource label.
- When initiating an investigation for an alert, a title, issue description, start time, and relevant resources for the alert are automatically pre-populated into the investigation creation panel.



Investigate an issue from an alert in Cloud Monitoring

- The alerting process helps you respond to issues when the performance of an application fails to meet acceptable values.
- The **Investigate** button appears for alerts that have a resource label, such as a Cloud Run service revision.

- When initiating an investigation for an alert, a title, issue description, start time, and relevant resources for the alert are automatically prepopulated into the investigation creation panel.

Click to zoom the image.

The Gemini Cloud Assist chat panel

The [Gemini Cloud Assist](#) chat panel in the Google Cloud console is another entry point from where you can initiate an investigation.

Setup

Before you can use Gemini Cloud Assist, you must perform the following setup steps:

Introduction

Gemini Cloud Assist can be set up for a Google Cloud project or folder.

For a project, these steps can be performed in the Google Cloud console using the gcloud CLI or the Google API.

To describe these steps, we'll use the gcloud CLI.

Click the Start or Next button (>) to learn more.

Step 2

Enable the Gemini Cloud Assist API

gcloud command:

```
$ gcloud services enable geminicloudassist.googleapis.com
```

Note: This also enables the Gemini for Google Cloud API, which is also required to use Gemini Cloud Assist.

Step 3

Grant IAM roles in a Google Cloud project

Grant the *roles/geminicloudassist.user*, *roles/recommender.viewer* roles to the user account:

gcloud commands:

```
$ gcloud projects add-iam-policy-binding PROJECT_ID --member=PRINCIPAL --  
role=roles/geminicloudassist.user
```

```
$ gcloud projects add-iam-policy-binding PROJECT_ID --member=PRINCIPAL --  
role=roles/recommender.viewer
```

Note: To get investigative assistance with products that the user doesn't already have access to, [additional IAM roles](#) might be needed.

Summary

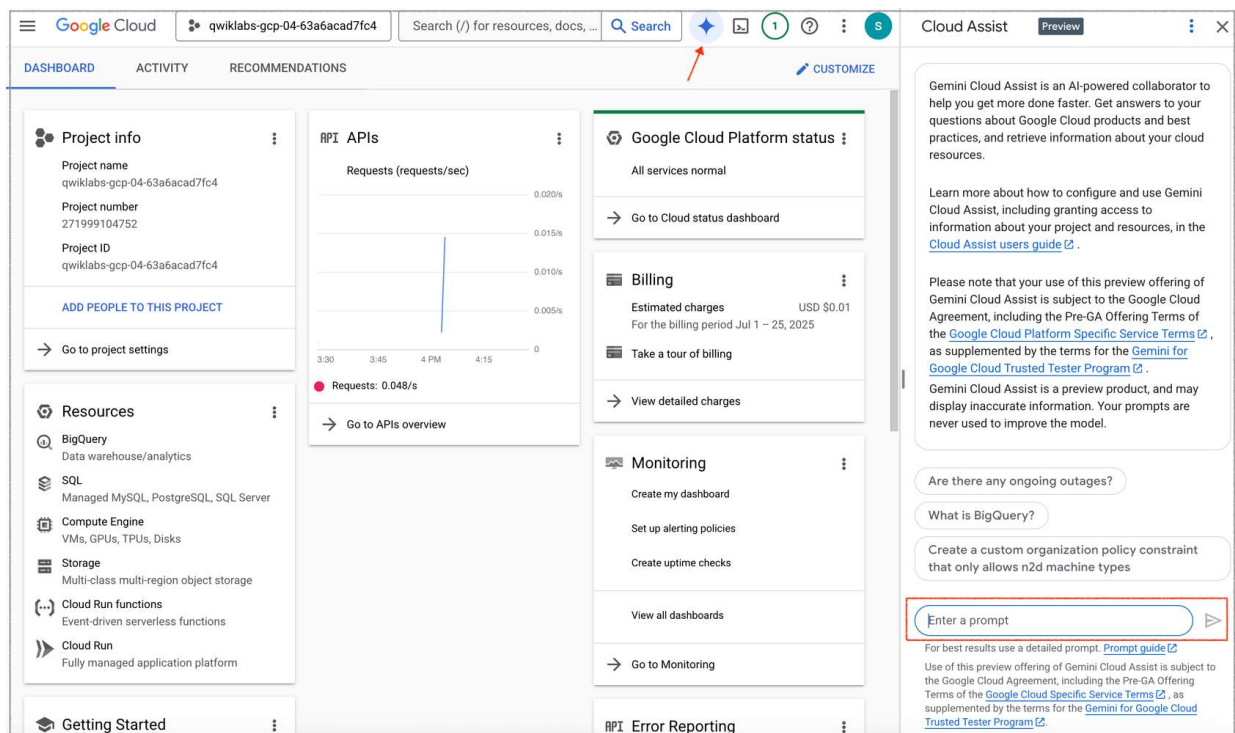
Gemini Cloud Assist is now ready to use in your project.

To [set up Gemini Cloud Assist for a folder](#), you can only use the Google Cloud console.

To learn about the pricing for Gemini Cloud Assist, visit the [documentation](#).

Access and use the chat panel

Now that you have set up the Google Cloud Assist chat panel for your project or folder, you can access it from any page in the Google Cloud console after selecting your project.



The screenshot displays the Google Cloud console interface for the project 'qwiklabs-gcp-04-63a6acad7fc4'. The top navigation bar includes the Google Cloud logo, project name, search bar, and a Gemini Cloud Assist icon (a blue diamond with a white 'G') highlighted by a red arrow. The main content area is divided into several sections: 'Project info' (Project name: qwiklabs-gcp-04-63a6acad7fc4, Project number: 271999104752, Project ID: qwiklabs-gcp-04-63a6acad7fc4), 'Resources' (BigQuery, SQL, Compute Engine, Storage, Cloud Run functions, Cloud Run), 'Getting Started', 'APIs' (Requests (requests/sec) graph), 'Google Cloud Platform status' (All services normal), 'Billing' (Estimated charges: USD \$0.01), 'Monitoring' (Create my dashboard, Set up alerting policies, Create uptime checks), and 'API Error Reporting'. On the right side, the 'Cloud Assist' chat panel is open, showing a 'Preview' button and a chat area with a text input field and a send button. The chat area contains text about Gemini Cloud Assist being an AI-powered collaborator and a preview product, along with links to the 'Prompt guide' and 'Gemini for Google Cloud Trusted Tester Program'.

Access the Gemini Cloud Assist chat panel

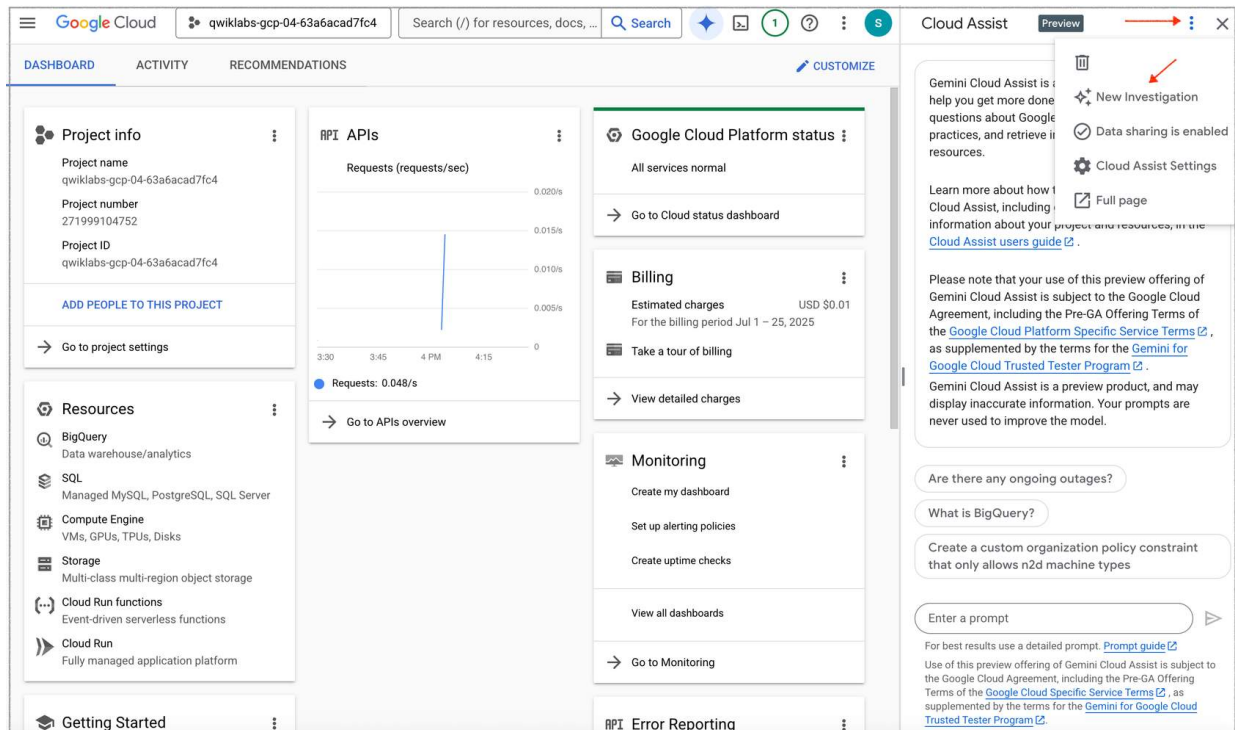
By writing statements or prompts in natural language in the chat panel, you can:

- Receive explanations, suggested actions, or guided workflows that help you to complete tasks quickly and efficiently.
- Access information on Google Cloud products and best practices for using them.
- Inspect your cloud resources, applications, and data.
- Get help with BigQuery datasets and databases.
- Investigate and troubleshoot issues with your Cloud resources and applications.

Click to zoom the image.

The New investigation option appears in the chat panel's drop-down menu.

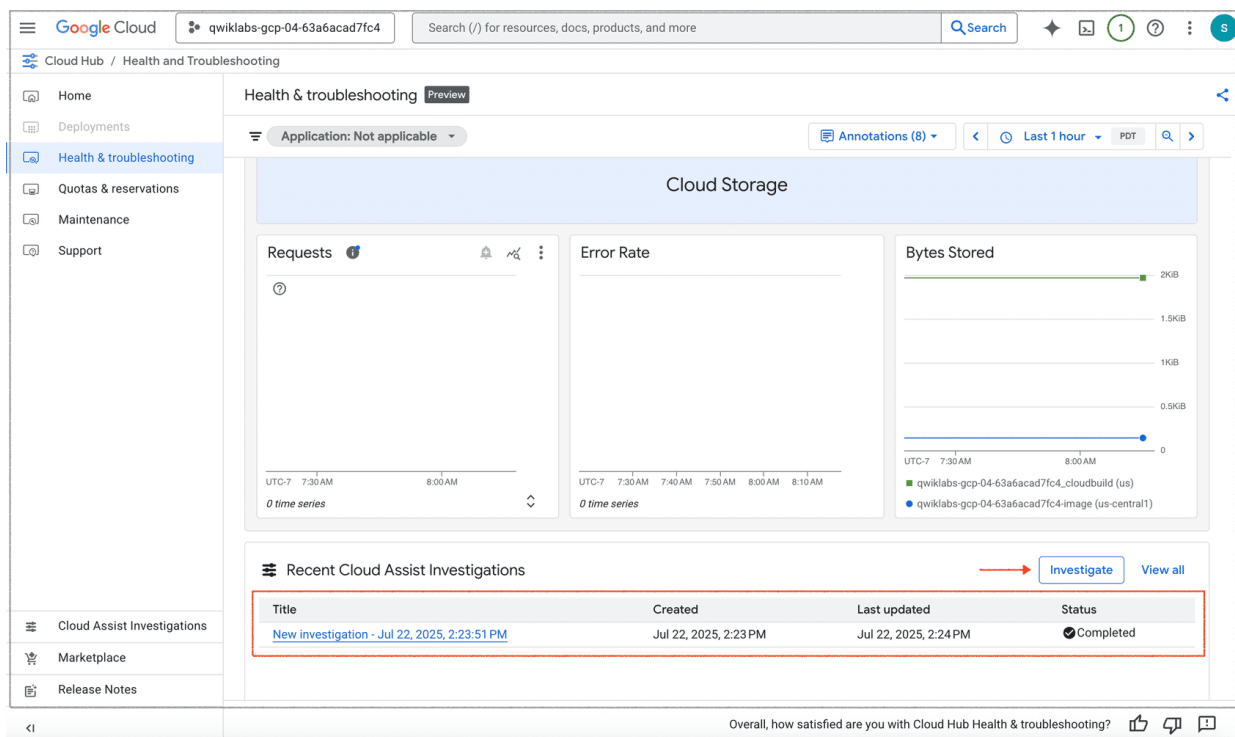
Click to zoom the image.



Create a new investigation from the Gemini Cloud Assist chat panel

Cloud Hub

Cloud Hub provides DevOps and SRE teams with a user interface in the Google Cloud console that contains a centralized view of operations data and insights for your applications and resources. Viewing this data together in one place helps you to take action on important issues, patterns, and trends.



Health and troubleshooting dashboard in Cloud Hub

You can:

- Access Cloud Hub from the products page in the console.
- View active alerts and events, including Google Cloud incidents.
- View your current investigations and create new ones in the health and troubleshooting dashboard.

Click to zoom the image.

Supported Google Cloud products

Cloud Assist Investigations supports the troubleshooting of many [Google Cloud products](#), including Cloud Storage, Cloud Run, and Google Kubernetes Engine.

- To investigate issues with your workloads that use these products, you can create investigations from product-related log messages within Logs Explorer.
- You can also create investigations from service logs in Cloud Run or container logs in GKE.

Knowledge check

You can create an investigation from an expanded log message in Logs Explorer for supported resources when the message has a severity level of _____ or higher.

Type your answer here

SUBMIT

Investigate and diagnose issues

The investigation details

In the previous module, we reviewed how to create an investigation in the Cloud Assist Investigations page in the Google Cloud console. Let's now review the details of that investigation.

The investigation details includes information about:

- The issue
- The observations
- The hypotheses and recommended fixes or additional troubleshooting steps.

Issue

Complete

Name	New investigation - Jul 16, 2025, 1:14:26 PM
Start time	July 16, 2025 at 12:44:26 PM UTC-7
End time	-
Project	apigee-edge-hm
Application	-
Error message	-
Issue description	Contents of a Cloud Storage bucket are not accessible.
Relevant resources (1)	
Storage buckets	clsaitools

Information about the issue in the investigation details

The issue

- Contains input information that was provided and used to generate the results of the investigation.
- Includes three types of resources that might be associated with the investigation:
 - User-provided resources: These are provided by the user or fetched from the initial error.
 - Discovered resources: Additional resources considered by Gemini while performing the investigation.
 - Relevant resources: The resources that Gemini considers relevant to the root cause hypothesis, which always include user-provided resources.

Click to zoom the image.

2 relevant observations

Collapse all

Bucket Access Restriction due to IAM Configuration

Config analysis7/16/25, 1:35PM

The Cloud Storage bucket 'clsaitools' has both 'uniformBucketLevelAccess' and 'bucketPolicyOnly' enabled, and 'publicAccessPrevention' is set to 'enforced'. This configuration restricts access to the bucket based solely on IAM policies, effectively preventing any access not explicitly granted through IAM. The 'lockedTime' for both 'uniformBucketLevelAccess' and 'bucketPolicyOnly' is set to '2025-06-05T23:12:07.589Z', which is in the past, meaning these configurations are active. Additionally, 'softDeletePolicy' is configured with 'retentionDurationSeconds' set to 604800 (7 days), and 'effectiveTime' is '2025-03-07T23:12:07.000Z'. This means that any deleted objects will be retained for 7 days, potentially impacting perceived accessibility if objects were recently deleted. Therefore, the combination of enforced public access prevention, bucket policy only, and uniform bucket-level access, along with the soft delete policy, is likely causing the inaccessibility of the bucket contents.

Relevant resources

clsaitools

Copy

Cloud Storage Access Logs Show No Errors

Logs

The Cloud Storage access logs do not show any errors related to accessing the bucket 'clsaitools' for object get or list operations. This suggests the issue might not be with the bucket itself, but with the application or service attempting access. `gcloud logging read resource.type="gcs_bucket" AND protoPayload.methodName="storage.objects.get" AND protoPayload.status.code!=0 --limit=10 --order=DESC` `gcloud logging read resource.type="gcs_bucket" AND protoPayload.methodName="storage.objects.list" AND protoPayload.status.code!=0 --limit=10 --order=DESC`

Copy

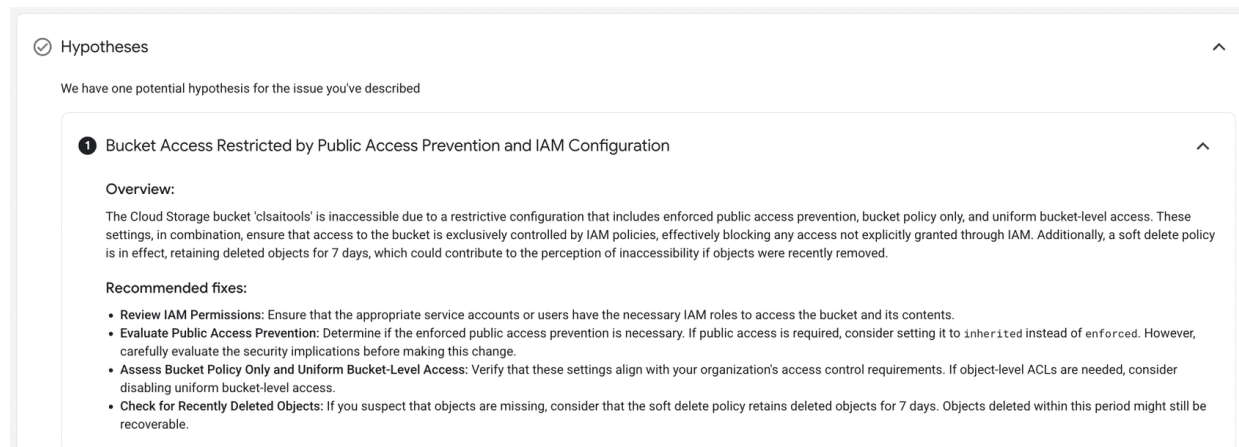
Observations generated by the investigation

The observations

- Generated by the investigation.

- Based on reviews of procedural runbooks, logs, outage messages, and alerts.
- Used to generate hypotheses on the root cause of the issue.

Click to zoom the image.



Hypotheses generated by the investigation on the root cause of the issue

The hypotheses

- Generated by the investigation.
- Based on the data collected and the observations made by the investigation.
- Contain an overview of what the issue could be.
- Provide recommended fixes that help you correct the issue, or
- Provide additional troubleshooting steps to confirm or refute the hypotheses.

Click to zoom the image.

Considerations

Here are some points to consider when using Cloud Assist Investigations to troubleshoot issues.

Click each button to expand the items and learn more.

Product API access —

As mentioned earlier, when you perform an investigation, Cloud Assist Investigations uses your credentials to access relevant product APIs and other information.

Each investigation creates and uses an [OAuth 2.0 token](#) with a grant scope of `API_CLOUD_PLATFORM` as part of its normal operation.

- The access provided by the OAuth 2.0 token is limited to the access available to the user or service account that initiates the investigation. This limits the investigation's access based on the access you have to the product APIs.
- The OAuth 2.0 token is not used for mutating data.

Data residency compliance —

You shouldn't perform investigations on data that is subject to residency or jurisdictional regulatory compliance requirements. The investigation resource that is created includes annotations and observations that can be stored in any Google Cloud data center.

Global resource analysis —

As part of an investigation, Cloud Assist Investigations analyzes global Google Cloud resources such as logs.

Investigations don't analyze:

- Location-specific content, such as data stored in regional log buckets.

- Data residing outside of Google Cloud.

Targeted troubleshooting —

Investigations are designed for targeted troubleshooting within Google Cloud environments.

An investigation is limited to diagnosing issues within:

- A single Google Cloud project
- A single application within App Hub

Timestamp usage —

Timestamps are a key piece of information in effective investigations. For the best results, verify that the start time reported in the investigation is reasonably accurate.

- An accurate start time enables time-based correlations to be made across seemingly unrelated events.

Resource usage —

To improve the outcome of an investigation, add the affected resources when creating the investigation.

You can only add resources that are in the project or application of the investigation's scope, and supported by Cloud Assist Investigations.

Dynamic results —

Investigation runs are dynamic, which means re-running the same investigation can produce results with small differences.

Some of the factors that contribute to differences between runs include:

- The probabilistic nature of output from large-language models such as Gemini.
- The non-static nature of the overall state of Google Cloud environments.

Diagnosing an issue from Logs Explorer

As mentioned earlier, the Logs Explorer in the Google Cloud console is another entry point from where you can initiate an investigation.

When you click on the Investigate button in the log message, the **Create new investigation** form opens. The form is prepopulated with information from the warning or error.

In this example, the warning is generated due to an authentication issue with a Cloud Run service. In this case, the resource identified is the Cloud Run service revision.

Google Cloud

qwiklabs-gcp

Logs Explorer

Project logs

Search a

All resources

All log names

Timeline

422 results

Actions

SEVERITY

TIME

> 2025-07-22 10:16:59.607

2025-07-22 10:17:10.814

Explain this log entry

```
{
  httpRequest: {10}
  insertId:
    "687fc796000dfb795bbc87f2"
  logName:
    "projects/qwiklabs-gcp-04-63a6acad7fc4/logs/run.googleapis.com%2Frequests"
}
```

student_04_b71e88f127c4@cloudshell:~

(qwiklabs-gcp-04-63a6acad7fc4) \$

Create new investigation

Describe your issue to start the investigation. Providing detailed information leads to a more accurate analysis.

Name

New investigation - Jul 22, 2025, 11:03:19 AM

Start time *

7/22/25, 10:17 AM

PDT

End time (optional)

PST

Defaults to 30 minutes prior, max 30 days ago

Time issue ended

Target project *

qwiklabs-gcp-04-63a6acad7fc4

Select

What type of issue are you facing?

☐ I want to describe what happened

Describe the issue you are encountering and include relevant Cloud resources

☒ I have an error message

Copy and paste verbatim errors to run a more accurate analysis

Error message

{\"textPayload\":\"The request was not authenticated. Either allow unauthenticated invocations or set the proper Authorization header. Read more at https://cloud.google.com/run/docs/securing/authenticating Additional troubleshooting documentation can be found at: https://cloud.google.com/run/docs/troubleshooting#unauthorized-

Enter the error messages verbatim for a more accurate analysis

Relevant project resources (optional)

+ Add resource

Delete

☐	Name ↑	Type	Location
☐	inventory-app-00001-jf7	Cloud Run revisions	us-central1

Create

Cancel

The Create new investigation form pre-populated with the log information.

Cloud Run Authentication Failure

Log theme 4 hours ago ^

The log entries indicate that requests to the Cloud Run service 'inventory-app' failed with a 403 status code because they were not authenticated. The service requires authentication, and the requests either didn't include the proper authorization header or the service is not configured to allow unauthenticated invocations. The requests originated from different sources, including a browser (Mozilla) and curl. The issue occurred on revision 'inventory-app-00001-jf7' in the 'us-central1' region.

Relevant resources [inventory-app-00001-jf7](#)

[Source Log Theme](#)

 Copy ▾

Observation indicating a Cloud Run authentication failure.

Observations

- The investigation generated an observation that indicates a Cloud Run authentication failure.
- This example indicates that the Cloud Run service requires authentication and the request did not include the proper credentials.

Click to zoom the image.

Hypothesis

- The investigation generated a hypothesis as to the root cause of the issue. In this case, an unauthenticated request was made to the service that required authentication. This produced a 403 error response from the service.
- Recommended fixes and related information are also provided to resolve the issue.

Click to zoom the image.

✓ Hypotheses

We have one potential hypothesis for the issue you've described

1 Cloud Run Service Requires Authentication and Request is Unauthenticated

Overview:

The Cloud Run service `inventory-app` is configured to require authentication, and the request you made was unauthenticated, resulting in a 403 error. The service `inventory-app` in the `us-central1` region, specifically revision `inventory-app-00001-jf7`, requires either allowing unauthenticated invocations or setting the proper Authorization header. The `HttpRequest` section of the log shows a 403 status code for a GET request to `https://inventory-app-271999104752.us-central1.run.app/`.

Recommended fixes:

- **Allow unauthenticated invocations:** If you want the service to be publicly accessible, you can configure it to allow unauthenticated invocations. You can do this using the `gcloud` command-line tool:

```
gcloud run services update inventory-app --region=us-central1 --allow-unauthenticated
```

Be aware that this will make your service publicly accessible.

- **Provide authentication credentials:** If you want to restrict access to the service, you need to provide authentication credentials with your request. This typically involves including an `Authorization` header with a valid identity token. You can obtain an identity token using various methods, such as the Google Cloud CLI or a service account.
- **Grant the `run.routes.invoke` permission:** Ensure that the IAM member used to generate the authorization token has the `run.routes.invoke` permission. You can grant this permission through the Cloud Run Admin and Cloud Run Invoker roles. If a service account invokes your service, ensure that the service account is a member of the Cloud Run service and has the Cloud Run Invoker role.

The hypothesis generated from the investigation and recommended fixes.

Knowledge check

Consider each factor when using Cloud Assist Investigations, and click the card to check your knowledge. Then click the arrows to navigate to the next or previous card.

How is access to product APIs granted?

To access product APIs when performing an investigation, an OAuth 2.0 token is created using the credentials of the user or service account that initiates the investigation.

Can I perform investigations on data that is subject to residency requirements?

You shouldn't perform investigations on data that is subject to residency or jurisdictional regulatory compliance requirements.

The investigation resource that is created includes annotations and observations that can be

Can I use investigations to analyze resources in a specific location?

Cloud Assist Investigations analyzes global resources, such as logs that reside in Google Cloud.

Investigations don't analyze location-specific content, such as data stored in regional log buckets.

What is the scope of a Cloud Assist investigation within a Google Cloud environment?

An investigation is scoped or limited to diagnosing issues within a single Google Cloud project or a single application within App Hub.

4 of 6

Why are timestamps important when creating investigations?

Timestamps are a key piece of information in effective investigations. Verify that the start time reported in the investigation is reasonably accurate.

An accurate start time enables time-based correlations to be made across seemingly unrelated events.

5 of 6

Why do results differ slightly between re-runs of the same

Investigation runs are dynamic, which means re-running the same investigation can produce results with small differences.

investigation?

This is partially due to the probabilistic nature of output from large-language models such as Gemini, and the non-

What did I walk away with?

You've reached the end of the course. We hope this course has enabled you to:

- ✓ Identify and navigate the Cloud Assist Investigations interface in the Google Cloud console.
- ✓ Utilize core features of Cloud Assist Investigations to investigate and diagnose common issues.
- ✓ Troubleshoot incidents with Cloud Assist Investigations.



Congratulations on completing this training!



